AID-DATA-008

PRODUCT MATERIAL STANDARD

Controlled material data for AI matching, filtering and product intelligence

1. Purpose

Defines how product materials are captured, normalized and separated from finishes in AIDOOi. The standard supports multi-material furniture without making the Universal Product Master unnecessarily complex.

2. Core Material Structure

Field	Requirement	Purpose
Primary Material Family	Required	Main material family that materially defines the product.
Primary Specific Material	Required when known	Precise material within the family.
Secondary Material(s)	Conditional	Other visually, structurally or functionally significant materials.
Secondary Specific Material(s)	Required when secondary material is known	Precise material for each secondary family.

3. Core Rule

Material Family provides the controlled AI/filtering layer. Specific Material preserves the real material detail. A product may contain multiple significant materials; these should be recorded rather than forcing the entire product into a single material label.

4. Controlled Material Family Vocabulary

Material Family	Typical Specific Materials
Wood	Oak, Walnut, Ash, Beech, Teak, Pine, Acacia
Engineered Wood	MDF, HDF, Plywood, Particleboard
Metal	Steel, Stainless Steel, Aluminium, Iron
Glass	Tempered Glass, Clear Glass, Smoked Glass, Frosted Glass
Leather	Full-Grain Leather, Top-Grain Leather, Genuine Leather, PU Leather
Fabric / Textile	Linen, Cotton, Velvet, Boucle, Polyester, Wool
Stone	Marble, Travertine, Granite, Quartzite
Engineered Stone	Quartz Composite, Sintered Stone
Plastic / Polymer	Polypropylene, Acrylic, Polycarbonate, ABS
Ceramic	Ceramic, Porcelain
Rattan / Wicker	Natural Rattan, Synthetic Rattan, Cane
Composite	Fiberglass, Wood-Plastic Composite or other verified composite
Foam / Padding	Polyurethane Foam, Memory Foam - when materially relevant
Other	Only for verified materials that cannot reasonably map to an approved family

5. Primary Material

Primary Material should represent the material that most strongly defines the product's physical identity, user interaction or visible construction. It is not necessarily the heaviest material by weight.

6. Secondary Materials

Record a Secondary Material when it is structurally, visually or functionally meaningful. Minor screws, hidden brackets, small fasteners or negligible internal components do not need to be captured as separate materials.

7. Multi-Material Example

Component	Material Family	Specific Material
Seat / arm sling	Leather	PU Leather or verified leather type
Frame	Metal	Specific metal if verified
Back cushions	Fabric / Textile	Specific textile if verified

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8. Material and Finish Must Stay Separate

Material answers what the product is made of. Finish answers how that material or surface is treated or presented. These concepts must not be merged.

Actual Construction	Material	Finish / Appearance
Solid walnut table	Wood - Walnut	Natural / Oil / Lacquer, if verified
MDF cabinet with walnut veneer	Engineered Wood - MDF	Walnut Veneer
Steel frame with antique brass appearance	Metal - Steel	Antique Brass Finish
MDF with marble-look surface	Engineered Wood - MDF	Marble-Look Finish

9. Important Anti-Misclassification Rule

Terms such as Walnut, Oak, Marble, Brass or Bronze may describe either a true material or only a finish/appearance. AIDOOi must not classify them as Specific Material unless the underlying construction is verified. For example, 'Walnut Finish' does not prove that the product is made from walnut wood.

10. Seller Terminology

Seller or manufacturer terminology should be normalized into the controlled Material Family while preserving verified Specific Material. Marketing phrases such as 'premium wood', 'luxury fabric' or 'marble effect' must not be treated as confirmed material composition.

11. Unknown / Unverified Material

Do not guess a precise material from appearance alone. When the family can be reliably identified but the exact material cannot, store the verified family and leave Specific Material unresolved. If even the family is uncertain, flag the material for review rather than inventing a value.

12. Variant Relationship

A material or upholstery choice may create a selectable Product Variant under the same Parent when the underlying model remains the same. A fundamental construction or model change may require a different Parent according to AID-DATA-002.

13. AI and Recommendation Rule

AIDOOi should use Material Family for robust filtering and broad preference matching, while Specific Material provides finer intelligence for texture, quality, maintenance, durability and user preference. Material data must remain independent from colour, style and finish.

14. Example - AIDOOi Test Lounge Chair

Field	Recommended Treatment
Primary Material	Use the dominant verified material based on final product evidence.
Leather component	Family = Leather; Specific = PU Leather / other type only if verified.
Frame	Family = Metal; Specific metal only if verified.
Back cushions	Family = Fabric / Textile; Specific textile only if verified.
Bronze / brass appearance	Treat as Finish unless the underlying metal itself is verified as bronze/brass.

15. Data Quality Rules

Rule	Standard
Use controlled families	Material Family must come from the approved vocabulary.
Preserve specificity	Store verified detailed composition in Specific Material.
Support multiple materials	Capture significant secondary materials instead of forcing one label.
Do not guess	Unknown precise materials remain unresolved.
Separate finish	Surface appearance or treatment must not overwrite actual material composition.
Avoid over-capture	Ignore negligible hardware/internal components unless functionally important.

16. Final Standard

Primary Material Family: required.
Primary Specific Material: required when known.
Secondary Material(s): conditional for meaningful additional materials.
Material vs Finish: always separate.
AI/filtering: driven primarily by controlled Material Families, enriched by verified Specific Materials.